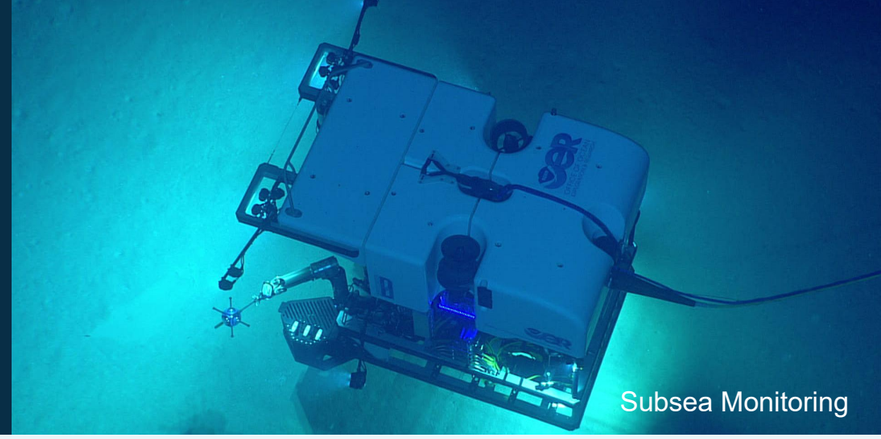
2026 National Undergraduate Embedded Chip and System Design Competition

# FPGA Multisource Aggregation and Coherent Optical Transmission

Underwater images and seismic data over one high-speed link

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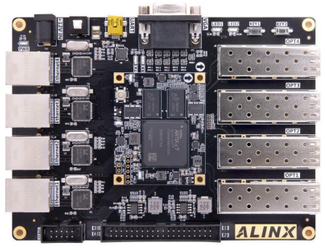

**331 km**
Unrepeatered G.654.E fiber link

**160 Gb/s**
10 WDM channels, aggregate gross rate

**4 Gbaud**
Dual-polarization QPSK

**BER < 0.01**
Best measured BER for each channel

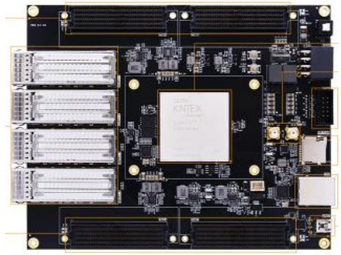
## Three FPGAs for Multisource Aggregation and Transmission




### AX7101  Data Source

Read image and seismic data

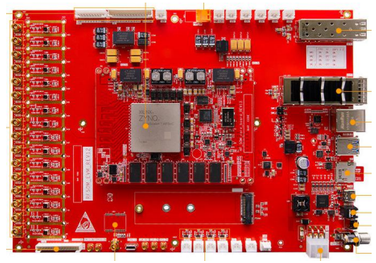
Object segments over UDP



### KU115  Aggregator

FIFOs + active-port scheduling

Cross-frame segments + CRC32



### RF47  Waveform TX

8-ch deskew / long PRBS / payload

DP-QPSK mapping + DAC output

### Coherent Optical TX / RX

IQ modulation → WDM → 331 km fiber

ICR → Oscilloscope → MATLAB DSP

CD compensation / equalization / decisions

CRC checks and object reassembly

## Full Recovery of Images and Seismic Data after 331 km

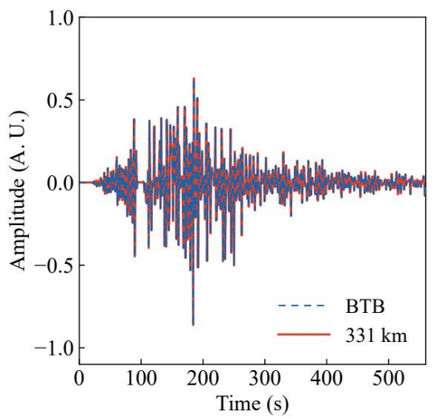
Measured Received Data



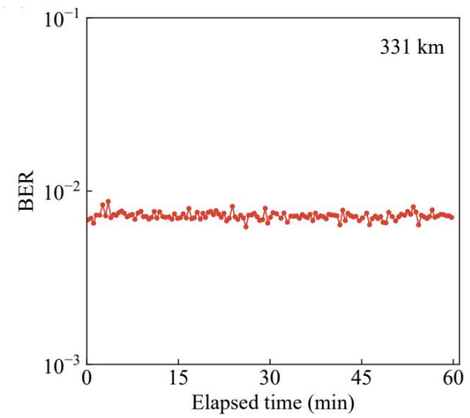
### 3 Underwater Images

160 × 120 RGB, reassembled by object offset

CRC-valid fragments only, with final object verification




### 11,200 Seismic Samples




### 1-hour Weighted BER  0.007193

130 valid windows. Pre-FEC BER below the 0.02 SD-FEC reference threshold.

Active-port aggregation with dynamic segmentation, verified by BER and end-to-end data integrity.

**FPGA Open Track, Junior Division**